



CANDIDATE NAME

CT GROUP

CENTRE NUMBER

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INDEX NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

26 August 2021

No additional materials are required.

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.

Answer **all** questions in the spaces provided on the question paper.

Shift
Laboratory

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 30
2	/ 25
Total	/ 55

QUESTION 1

Plant cells contain enzymes which catalyse some of their metabolic reactions. Some of these enzymes catalyse the release of oxygen from hydrogen peroxide.

A cylinder of potato tissue will have these enzymes on the surface.

When hydrogen peroxide solution and a cylinder of potato tissue are mixed, oxygen bubbles are released.

You will need to investigate the effect of surface area by:

- changing the surface area
- counting the number of bubbles of oxygen released in a set time.

You are provided with the materials shown in Table 1.1 and Table 1.2.

Table 1.1

labelled	contents	hazard	volume / cm ³
H	hydrogen peroxide solution	moderate	40
W	water	none	100

If any of **H** comes into contact with your skin, wash off immediately under cold water.

It is recommended that you wear suitable eye protection and gloves.

Table 1.2

labelled	contents	details	quantity
P	potato cylinders	same cross-sectional area	4

- (a) To investigate the effect of surface area, other variables need to be standardised.

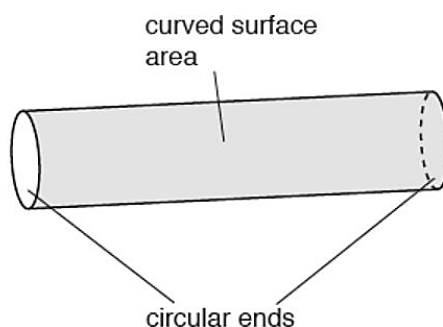
Each potato cylinder has been provided with the same diameter but with **different** lengths.

Each cylinder of potato tissue must be cut to the **same** length.

Proceed as follows.

- 1** Cut each of the four potato cylinders in the beaker labelled **P**, to a length of 20 mm.

To investigate the effect of surface area, the surface area can be changed by cutting each of these four cylinders into a different number of pieces.



- (i) Measure the diameter of **one** of the cylinders in **P** and calculate the curved surface area, to the nearest whole number. (use π as 3.14)

curved surface area = _____ [2]

- (ii) For the potato cylinders in **P**, calculate the area of one circular end to the nearest whole number. (use π as 3.14)

area of one circular end = _____ [1]

- (iii) Complete Table 1.3 to calculate the total surface area when using different numbers of pieces of potato cylinders.

Table 1.3

number of pieces cut from one cylinder	number of circular ends	area of one circular end from (a)(ii) / mm ²	surface area of all the ends / mm ²	curved surface area from (a)(i) / mm ²	total surface area / mm ²
1	2				
2	4				

[2]

- 2 Cut each of the four cylinders into the number of pieces, as shown in Table 1.3.
- 3 Put the pieces into the shallow dish labelled C.
Cover with a damp paper towel to prevent the pieces from drying out.

You will use the apparatus as shown in Fig. 1.1.

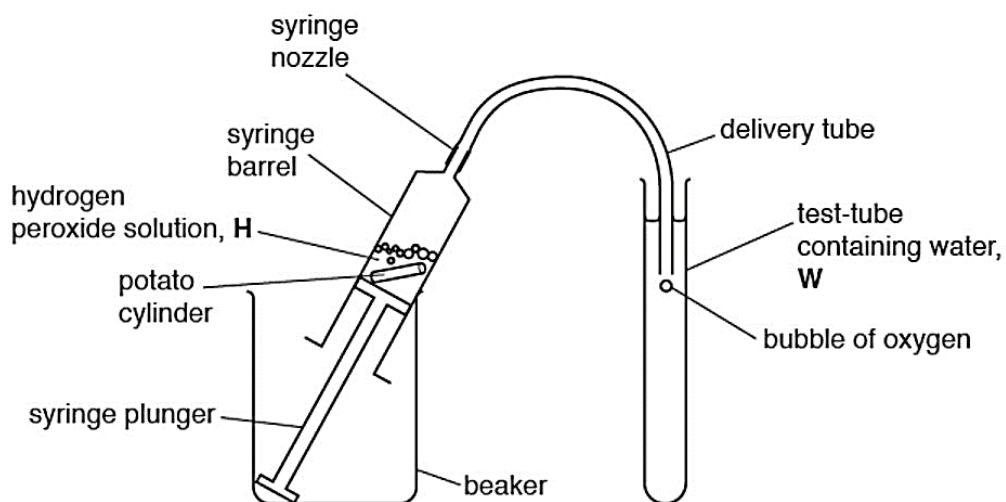


Fig. 1.1

The position of the delivery tube in the test-tube of water should be standardised to have confidence in the results.

- (iv) Describe how you will standardise the position of the delivery tube in the test-tube of water, as shown in Fig. 1.1.

.....

.....

..... [1]

- 4 Set up the test-tube with **W** and the delivery tube, as shown in Fig. 1.1 and described in (a)(iv). Do **not** attach the tubing to the syringe. You may stand the test-tube in the test-tube rack provided.
- 5 Remove the plunger from the 10 cm³ syringe.
- 6 Put potato tissue into the barrel of the syringe, for example the whole cylinder in one piece.
- 7 Replace the plunger and push it to the 4 cm³ mark, as shown in Fig. 1.2.

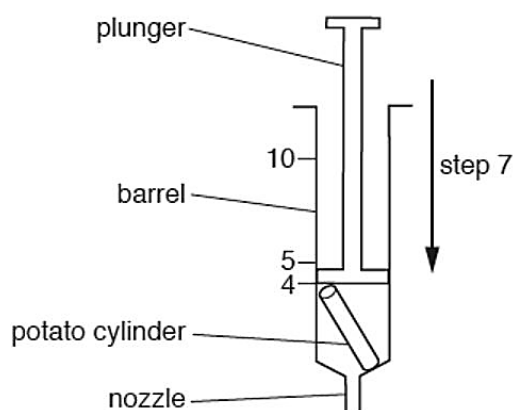


Fig. 1.2

- 8 Put the nozzle of the syringe into the plastic vial containing **H**.

- 9 Pull the plunger out to the 10 cm³ mark so that **H** enters the syringe, as shown in Fig. 1.3.

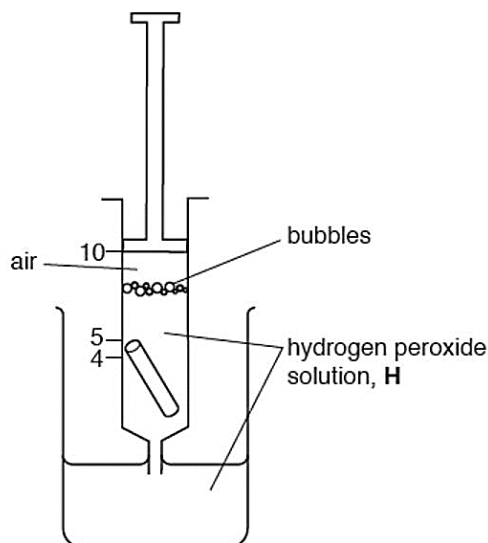


Fig. 1.3

- 10 Hold the syringe above the vial containing **H** and push the plunger to adjust the level of **H** to the 5 cm³ mark in the syringe, as shown in Fig. 1.4.

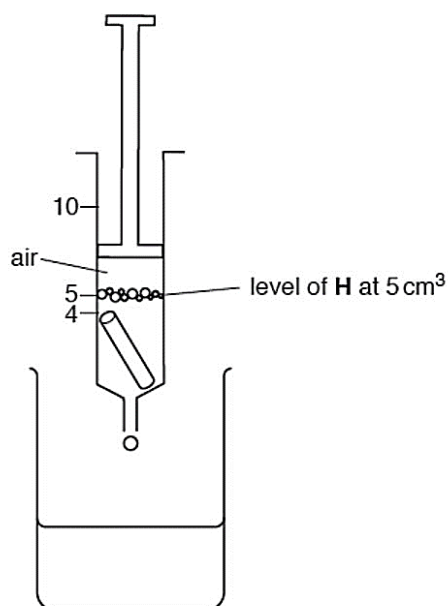


Fig. 1.4

- 11 Turn the syringe upside down so that the nozzle is up and there is air in the top of the syringe barrel. Carefully wipe the nozzle with a paper towel to remove excess **H**.
- 12 Tap the syringe barrel to make sure all the potato pieces are in **H**.

- 13 Attach the delivery tube to the nozzle to make an airtight fit.
 - 14 Put the syringe into a beaker as shown in Fig. 1.1 (page 4).
 - 15 Put the end of the delivery tube back into the test-tube as described in **(a)(iv)**.
 - 16 Start timing when the first bubble is observed in the water in the test-tube.
 - 17 Count the bubbles at intervals of 30 seconds up to 120 seconds. Record the results in **(a)(v)**.
 - 18 Using a paper towel to avoid **H** coming into contact with your skin, remove the delivery tube from the syringe, keeping the syringe nozzle up.
 - 19 Then push the plunger to empty as much as possible of **H** into the container labelled '**For waste**'.
 - 20 Slowly pull out the plunger and put the potato tissue and remaining **H** into the container labelled '**For waste**'.
 - 21 Repeat step 6 to step 20 with both of the pieces from the cylinder cut into two pieces.
 - 22 Repeat step 6 to step 20 with each of the other two cylinders which have been cut into different number of pieces.
- (v)** Record your results for the total surface area as shown in Table 1.3 **and** the number of bubbles at each 30 seconds in an appropriate table.

(vi) Using the results in (a)(v), calculate for the **largest** surface area:

- the mean number of bubbles in 30 seconds
- the rate of activity, number min^{-1} .

Show all the steps in your working and use appropriate units.

mean number of bubbles = _____

rate of activity = _____

[3]

(vii) Complete Table 1.4 to suggest:

- **one** other significant source of error in this procedure
- how to make improvements to reduce these errors.

Table 1.4

significant source of error	how to make an improvement
different sizes of bubbles released	
another significant source of error	

[3]

- (b)** The water potential of potato tissue can be determined using different methods. One such method involves the use of a range of concentrations of mannitol.

Mannitol is a sugar-like molecule which is not absorbed by cells.

You are provided with a stock solution of mannitol. The concentration of the stock solution is 1.0 mol dm^{-3} .

You are required to make a simple dilution from the stock solution to prepare a range of concentrations of mannitol that you can use to find the water potential of potato tissue.

- (i)** State 5 suitable concentrations of mannitol you will prepare from the stock solution.

..... [1]

- the 5 concentrations of mannitol prepared in **(b)(i)**, (*details of simple dilution **not** required*)
- potato tissue,
- weighing balance.

- cork borer,
- normal laboratory glassware, e.g. beakers, measuring cylinders, test-tubes, etc.,
- mm ruler,
- syringes.

[5]

- (iii) Suggest how the results obtained from your experiment in (b)(ii) can be used to estimate the water potential of potato tissue.

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.....

.....

..... [2]

- (c) Scientists use another method to investigate the determine the water potential of potato tissue. This was done by placing potato cylinders in solutions with different water potentials for 20 hours. All other variables were kept constant.

The tissue was then compressed. The scientists measured the sound that the potato tissue made as it was compressed using an acoustic emission meter.

The measurement of acoustic emission from compressed potato tissue can be used to judge the quality of the potato.

The results are shown in Table 1.5.

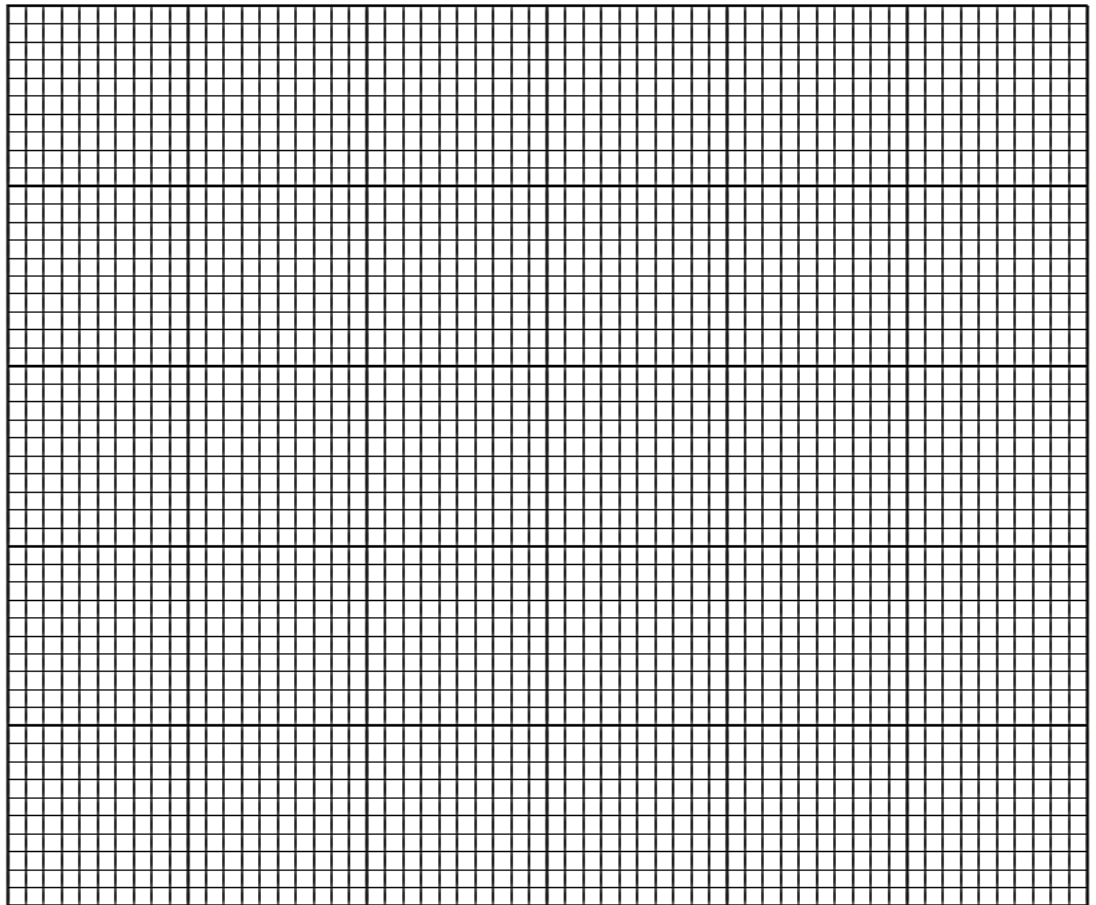
Table 1.5

water potential / MPa	acoustic emission / arbitrary units
-0.3	95
-0.5	76
-0.6	68
-0.9	50
-1.2	43
-1.5	24

- (i) Plot a graph of the data in Table 1.5 on the grid provided.

The position of zero on the x-axis is shown.

Use a sharp pencil for drawing graphs.



0

[4]

- (ii) Use your graph in **(b)(i)** to estimate the water potential of the potato tissue at an acoustic emission value of 80 arbitrary units.

water potential = _____ [1]

[Total: 30]

QUESTION 2

During this question you will require access to a microscope, a stage micrometer and slide **R1**.

R1 is a slide of a stained transverse section through the stem of a rice plant. The centre of the stem is hollow.

You are not expected to be familiar with this specimen.

Use a sharp pencil for drawing.

- (a) Use your microscope to observe the different tissues in the region of slide **R1** shown by the shaded area in Fig. 2.1.

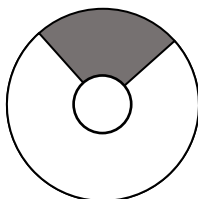


Fig. 2.1

- (i) Use the space provided to draw a large plan diagram of the sector of the stem on **R1**, as shown by the shaded area in Fig. 2.1.

A plan diagram shows the arrangement of different tissues, including their correct shapes and proportions. No cells should be drawn.

Use **one** ruled label line and the label **A** to identify an air space.

Within this sector of the stem shown by the shaded area in Fig. 2.1, there are number of air spaces.

You are required to investigate the proportion of the stem occupied by one air space. The ratio of the mean diameter of the air space compared to the mean thickness of the sector can be calculated.

- (ii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, describe how you can determine the diameter of an air space.

.....

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..... [2]

- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the mean diameter of an air space, in μm .

Show the measurements that you made and your working.

[3]

- (iv) The mean thickness of the sector has been determined to be 0.98 mm.

Calculate the ratio of the mean diameter of the air space to the mean thickness of the sector.

[2]

- (v) Within this sector of the stem there is a number of air spaces. Observe the cells that are found between two air spaces.

Select a group of **four** adjacent cells that are found between two air spaces. Each cell must touch at least two other cells.

Make a large labelled drawing of this group of four cells.

[4]

Fig. 2.2 is a photomicrograph of a stained transverse section through a stem of a different type of plant.

You are not expected to be familiar with this specimen.

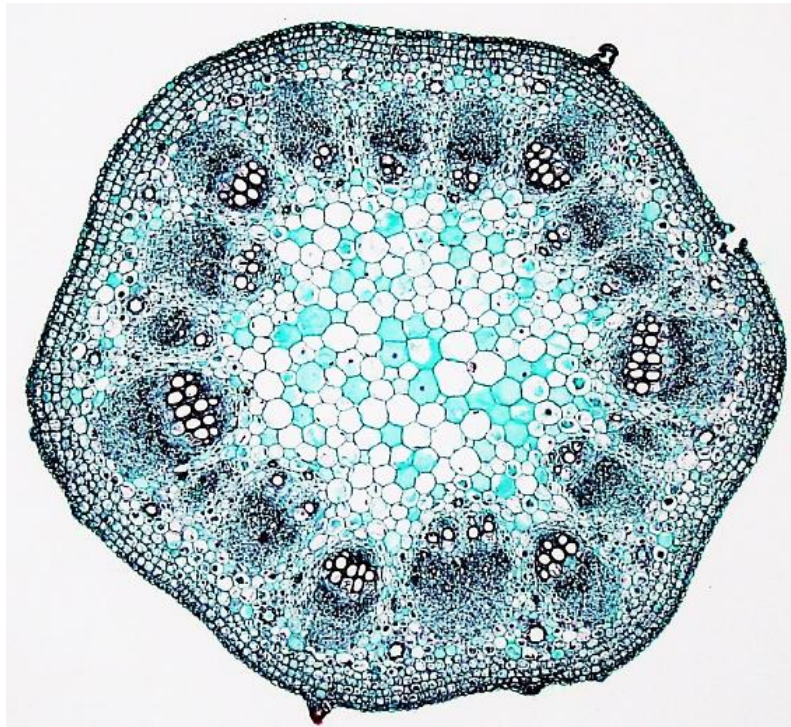


Fig. 2.2

(vi) Identify observable differences between the stem on **R1** and the stem in Fig. 2.2.

Record the observable differences in Table 2.1.

Table 2.1

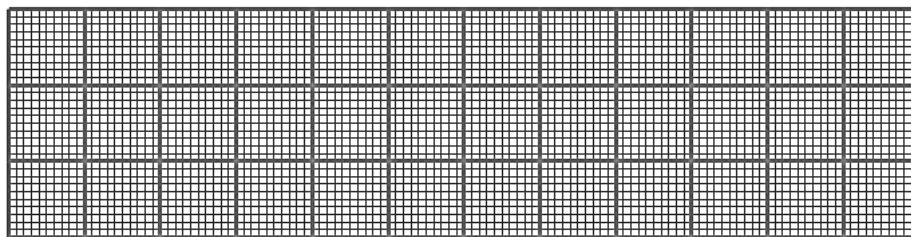
feature	R1	Fig. 2.2

[2]

- (b) You are provided with a plastic vial, labelled **R**, containing grains of rice. The rice grains were randomly chosen from the F_2 generation of a cross between a variety of long-grained rice and a variety of short-grained rice. The length of rice grains is controlled by many minor genes and one major gene with two alleles.

Proceed as follows.

- 1 Select **30** grains of rice randomly. Discard any grains that are not whole.
- 2 Lay each of the 30 grains of rice on the 1-mm graph grid. Measure the length of each grain to the nearest mm.



- (i) Use the space provided below to record your results in a tally chart for a range of lengths from 1 mm to 12 mm, at 1-mm intervals.

Each occurrence of a length is recorded by a tally mark (a single vertical line) and every fifth tally is drawn horizontally across the four vertical tally marks. The tallies can then be counted to give the frequency.

[2]

- (ii) Explain why all the long grains of rice are not exactly the same length.

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.....

[2]

- (c) In a more extensive study of this F₂ generation, the data in Table 2.2 were obtained.

Table 2.2

short grain	long grain
1436	4470

A χ^2 (chi-squared) test was carried out using the formula below, creating Table 2.3.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

key to symbols:

Σ = 'sum of

O = observed 'value'

E = expected 'value'

Table 2.3

observed frequency (O)	expected frequency (E)	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
1436	1476.5	- 40.5	1640.25	
4470	4429.5	40.5	1640.25	
$\chi^2 = \sum \frac{(O - E)^2}{E}$				

- (i) Complete the final column of Table 2.3.

[1]

- (ii) Using the calculated χ^2 value from (c)(i) and Table 2.4, find the probability that the observed results deviated from the expected by chance. Explain how this probability is determined.

Table 2.4

degrees of freedom	probability						
	0.5	0.20	0.10	0.05	0.02	0.01	0.001
1	0.46	1.64	2.71	3.84	5.41	6.64	10.83
2	1.39	3.22	4.61	5.99	7.82	9.21	13.82
3	2.37	4.64	6.25	7.82	9.84	11.35	16.27
4	3.36	5.59	7.78	9.49	11.67	13.28	18.47

probability = _____

explanation:

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.....

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[2]

- (iii) State what conclusion may be drawn from the probability found in (c)(ii).

.....

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[1]

[Total: 25]

--- END OF PAPER---

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